DiffRSMT: A Diffusion-Driven Interactive Multitask Learning Model for Joint Remote Sensing Image Segmentation, Height Estimation, and Boundary Detection

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***Abstract*—Remote sensing (RS) multitask learning (MTL) aims to jointly perform several interpretation tasks within a unified framework. However, existing MTL methods often treat these tasks as individual problems, but overlook the importance of task interactions. In this article, we propose a novel diffusion-driven MTL framework, named DiffRSMT, for joint RS image segmentation, height estimation, and boundary detection. Unlike existing approaches that require separate diffusion processes for each task, DiffRSMT deploys a unified diffusion framework, which consists of two main components: a label-unified encoding (LuE) module and a task-interactive decoding module. Specifically, the LuE module aggregates the heterogeneous task labels into a unified latent representation, enabling the network to effectively capture task-interactive features. The task-interactive decoding module employs a time-conditioned attention mechanism during the denoising phase to enhance the interactions between tasks, which can improve the model's MTL capability. In addition, a time-conditioned residual convolution module is introduced to further refine the denoising process. Extensive experiments on multiple RS datasets demonstrate that DiffRSMT provides better performance than several state-of-the-art MTL methods. The code is available at https://github.com/sycs-2024/DiffRSMT**

***Index Terms*—Boundary detection, diffusion model, height estimation, multitask learning (MTL), remote sensing (RS), semantic segmentation.**

I. Introduction

REMOTE sensing (RS) imagery interpretation aims to extract meaningful information from RS data, such as semantic segmentation [1], [2], boundary detection [3], and height estimation [4]. It plays a vital role in Earth observation and has been widely applied in various practical applications, such as environmental monitoring, urban planning, and disaster response. With the rapid advancement of artificial intelligence, a variety of deep-learning (DL)-based methods have been developed for RS image interpretation [5], [6], [7].

With the increasing number of RS image interpretation tasks being addressed, artificial intelligence-based interpretation methods can be categorized into two classes: single-task and multitask learning (MTL) approaches [8]. Single-task models are typically designed to solve a specific problem. While they often achieve competitive performance on individual tasks, their limited scope may lead to inefficient

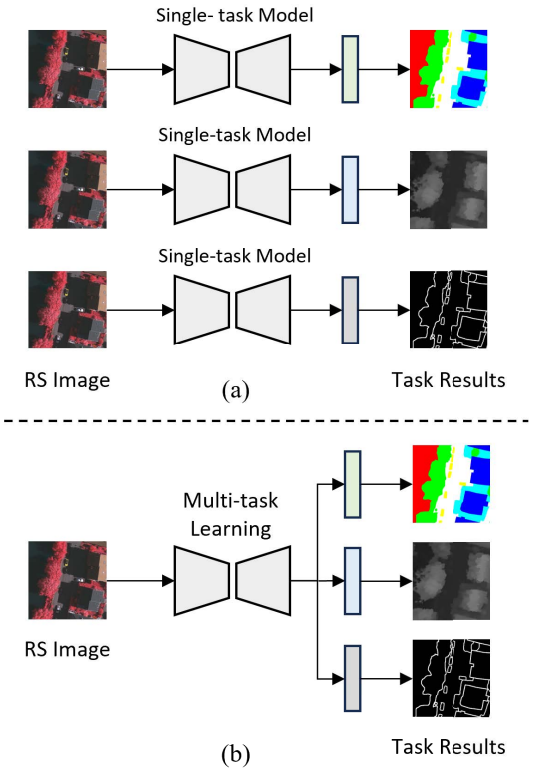


Fig. 1. Comparison between single-task learning and MTL paradigms. MTL enables shared representation learning while maintaining task-specific outputs. (a) Single-task learning paradigm. (b) MTL paradigm.

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use of available information and computational resources. In contrast, MTL can simultaneously address multiple related tasks within a unified model. Compared with single-task learning, MTL-based methods offer two key advantages [9]. First, they improve efficiency by allowing tasks to share network parameters. Second, they enable more effective representation of features among tasks. As shown in Fig. 1, the single-task models are trained individually with no information exchange, while MTL-based models can learn the shared features among tasks, which can benefit each task through joint learning.

Owing to the ability of automatic feature extraction, DL-based methods can learn rich representations from RS images without the need for handcrafted features [10]. According to the position of task features shared, DL-based MTL methods can be divided into two categories: encoder- and decoder-based models. Encoder-based methods learn the shared features across tasks in the early stage of a network. With a common backbone or shared feature extractor, these methods can capture general semantic and spatial features that are useful for all tasks. Typical methods include LB-MTL [11] and RSCoTr [12]. Decoder-based methods emphasize feature sharing in the decoding stages of the network [13]. For instance, to effectively explore the features among tasks, [14] proposed a multitask network that integrates a Transformer and an attention mechanism to enhance the interactions between decoders.

More recently, generalist models have emerged as a new paradigm for MTL [15], [16]. Equipped with large backbones, these models usually adopt a task-agnostic design by formulating different tasks under a shared modeling framework. A main class of such models is foundation models, which are typically pretrained on large-scale RS datasets. For example, SkySense [17], a typical foundation model, combines multiple backbones to perform different RS interpretation tasks. Similarly, a recently proposed foundation model, named MTP [18], is developed with a multitask training strategy. However, these approaches often rely heavily on an encoder to capture shared features, while overlooking task interactions that are crucial for effective MTL.

Generative models, as another category of generalist models, have recently gained much attention [19]. Unlike discriminative models, generative models aim to model the underlying data distribution and generate outputs through learned probabilistic processes. Among them, diffusion models have demonstrated potential in a variety of RS image interpretation tasks [20]. For instance, a diffusion-based model [21] is proposed for ship instance segmentation using SAR data. Based on latent diffusion principles, a diffusion model and Transformer-integrated network [22] are proposed for heterogeneous feature fusion in multisource RS classification.

Although diffusion models have shown promising results in several RS interpretation tasks, most RS diffusion studies focus on individual tasks, such as segmentation, restoration, or classification. Recent diffusion-based MTL methods further show the potential of diffusion models for general dense prediction. However, these methods are not specifically designed for RS multitask interpretation with heterogeneous labels, where semantic masks are categorical, height maps are continuous, and boundary maps contain thin spatial structures. Therefore, how to represent these heterogeneous labels in a compact shared latent space and model task interactions during conditional denoising remains challenging.

To address the above challenges, in this article, we propose a diffusion-driven MTL method, named DiffRSMT, for RS image multitask interpretation. Unlike previous methods that apply an individual diffusion process for each task, our model integrates the diffusion processes of different tasks into a unified framework. The proposed DiffRSMT consists of two main components: a label-unified encoding (LuE) module and a task-interactive decoding module. In the LuE module, different task labels are encoded into a unified latent representation via a label embedding layer, converting the heterogeneous task labels into a unified feature map. In the task-interactive decoding module, a time-conditioned task-interactive attention (TCTIA) mechanism is designed during the denoising phase, which learns to use task representations to collaboratively query the unified latent representation and thus enhance the task-interactive representations. To evaluate the effectiveness of our proposed DiffRSMT, three RS image interpretation tasks, including semantic segmentation, height estimation, and boundary detection, are conducted in the experiments. The main contributions of this article are summarized as follows.

1) A diffusion-driven MTL model, named DiffRSMT, is proposed for RS image multitask interpretation. Instead of constructing an independent diffusion process for each task, DiffRSMT jointly performs semantic segmentation, height estimation, and boundary detection within a unified diffusion framework.
2) The proposed DiffRSMT consists of two novel modules, that is, the LuE module and the task-interactive decoding module. The former learns to aggregate embedding features from multiple task labels into a unified latent representation, while the latter proposes a time-conditioned mechanism to capture and enhance task-interactive representations during the denoising process, enabling more effective MTL.
3) The effectiveness of DiffRSMT is validated by jointly performing semantic segmentation, height estimation, and boundary detection on multiple RS datasets. Experimental results demonstrate that it outperforms several state-of-the-art methods.

The remainder of this article is organized as follows. Section II reviews related works on MTL and diffusion models. Section III presents the details of the proposed DiffRSMT. Section IV reports the experimental results and analysis. Finally, Section V concludes this article and discusses future directions.

II. Related Works

A. Multitask Learning

RS MTL aims to simultaneously perform multiple interpretation tasks within a unified framework [8]. By enabling feature sharing among tasks, MTL can improve learning efficiency and enhance the acquisition of general representations [23].

In recent years, DL-based methods have been widely used for RS image MTL. For instance, [24] proposed a multitask

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network for interpretation tasks of time series RS images. By efficiently enhancing the representation of spatiotemporal differences, the method improves the detection of change areas and change moments. To improve the accuracy of recognition and vectorization of agricultural cultivation field parcels, [25] proposed a multitask framework that combines region, boundary, and distance tasks with a discrete cosine transform module. Recent advances have introduced generalist models into the MTL paradigm. These models employ large-scale pretraining and high-parameter networks to support diverse interpretation tasks. For instance, [26] proposed a unified multitask fine-tuned foundation model, which utilizes a shared encoder and separate decoders for individual interpretation tasks. In addition, [18] proposed a multitask pretraining model, which conducts multitask model training on the segment anything (SAM)-annotated dataset and RS segmentation dataset, encompassing semantic segmentation, instance segmentation, and rotated object detection.

However, many of these models rely heavily on an encoder for feature sharing, while often overlook the importance of task interactions in the decoding process.

B. Diffusion Model

A diffusion model [27] is proposed to generate data by iteratively denoising a random noise signal through a Markovian process, gradually transforming pure noise into meaningful data representations. Compared with other generative models such as generative adversarial networks (GANs) [28], the diffusion model enables the network to capture complex data distributions, making it particularly useful for tasks such as image generation and restoration [29].

Recently, owing to their ability to model complex data distributions, diffusion models have been applied to several RS image interpretation tasks, such as RS image super-resolution [30] and semantic segmentation [29]. For instance, DiffSAR-ShipInst [31] formulates SAR ship instance segmentation as a diffusion-based denoising process from noisy boxes to target boxes. A conditional diffusion model is also used for crop field aerial image super-resolution [32]. These studies show the potential of diffusion models for RS interpretation. However, most diffusion-based RS methods are still designed for individual tasks and do not explicitly model interactions among heterogeneous dense prediction targets.

Diffusion-based multitask dense prediction has also been studied in general computer vision. For instance, Diffusion-MTL [23] is proposed to reformulate partially annotated multitask dense prediction as a pixel-level denoising problem. It introduces a joint diffusion and denoising paradigm with multitask conditioning to rectify noisy task predictions under partial labels. TaskDiffusion [33] further investigates fully supervised multitask dense prediction with diffusion models, which encodes task-specific labels into a task-integration feature space, and thus models task relations during joint denoising. These studies demonstrate the potential of diffusion models for jointly modeling multiple dense prediction tasks, but they are mainly designed for general visual scenes and task settings.

Different from these general vision methods, this work focuses on RS multitask interpretation with heterogeneous labels. RS images have complex spatial structures and scale variations, which require task interaction to be closely coupled with image-derived features. Therefore, we propose DiffRSMT, a novel diffusion-driven framework for RS multitask prediction. The proposed LuE module maps heterogeneous RS task labels into a shared task-label latent space. The task-interactive decoder then uses shared image features and initial task predictions as conditions to exchange task information during denoising. Experimental results conducted on several RS datasets demonstrate that DiffRSMT provides better performance when compared with several state-of-the-art MTL methods.

III. Methodology

A. Preliminary of Diffusion Model

Diffusion models [27] are a class of generative models that learn a data distribution by reversing a gradual noising process. They consist of two main stages: a forward diffusion process that incrementally adds Gaussian noise to an input data sample, and a learned backward denoising process that iteratively removes the noise to recover the original data from a noise latent.

The forward diffusion process, denoted as q, gradually corrupts an initial sample x_0 over T discrete timesteps. At each timestep t, a small amount of Gaussian noise is added according to a predefined variance schedule $\{\beta_t \in (0,1)\}_{t=1}^{T}$. This forms a Markov chain where each subsequent state x_t is sampled from a conditional Gaussian distribution

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t;\ \sqrt{1-\beta_t}x_{t-1}, \beta_t \mathbf{I}\right). \tag{1}$$

A key property of this process is that we can sample x_t at any arbitrary timestep t directly from the original image x_0 in a closed form. By defining $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$, the distribution of x_t conditioned on x_0 is given by

$$q(x_t|x_0) = \mathcal{N}\left(x_t;\ \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\mathbf{I}\right). \tag{2}$$

As t approaches T, the distribution of x_T converges to a standard isotropic Gaussian distribution.

The backward denoising process, denoted as p_θ, aims to learn the true posterior distribution $q(x_{t-1}|x_t)$ to reverse the diffusion. Starting from pure noise $x_T \sim \mathcal{N}(0, \mathbf{I})$, the model learns to iteratively denoise it step-by-step to generate a clean sample $\tilde{x}_0$. The process can be expressed as

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \tag{3}$$

where θ represents the parameters to be optimized [34].

For RS image prediction tasks, the reverse process can be conditioned on the input image and used to recover task-specific outputs from a noisy latent representation.

B. Architecture Overview

The proposed DiffRSMT is a diffusion-driven framework designed to perform multiple RS image interpretation tasks simultaneously. As illustrated in Fig. 2, the architecture takes an RS image as input and jointly generates outputs for semantic segmentation, height estimation, and boundary detection. Unlike previous diffusion-based methods that require separate

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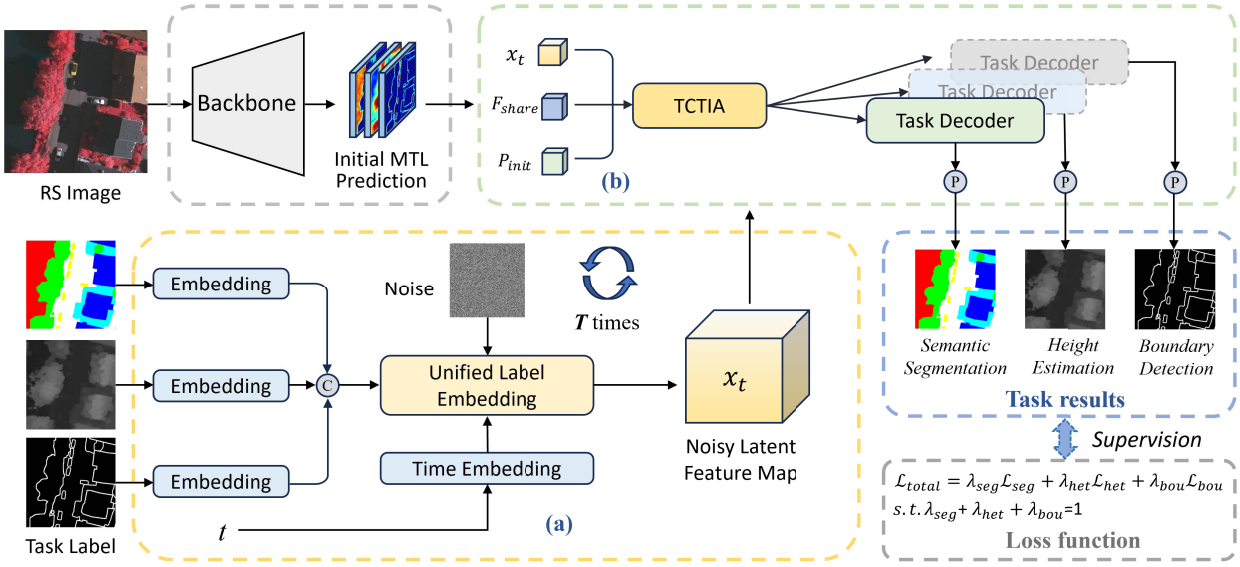


Fig. 2. Overall architecture of the proposed DiffRSMT framework. An input RS image is first processed by a backbone to extract shared features and an initial multitask prediction. In the LuE module, task labels are embedded into a unified representation and subjected to a forward diffusion process. The task-interactive decoding module then learns to reverse this process, conditioned on the image features and timestep. (a) LuE. (b) Task-interactive denoising.

models for each task, DiffRSMT integrates the entire MTL process into an end-to-end denoising diffusion pipeline.

Our proposed method adopts ConvNeXt as the backbone. Given an input RS image, denoted as $I \in \mathbb{R}^{H\times W\times C}$, the backbone extracts a shared feature representation F_{shared} from I. To enrich the conditioning information for the diffusion process, an initial multitask prediction from F_{shared} is also generated, denoted as P^k_{init}, where K is the number of tasks. The initial predictions are concatenated to form a more general representation, denoted as P_{init}. Then this initial prediction is fused with the shared features to form a composite conditional representation, which will be used in the subsequent modules.

The proposed DiffRSMT consists of an LuE module and a task-interactive decoding module. First, the LuE module maps heterogeneous task labels, including semantic segmentation, height estimation, and boundary detection maps, into a shared latent representation x_0. The forward diffusion process then progressively adds Gaussian noise to x_0 over T timesteps, producing the noisy latent representation x_t. The task-interactive decoding module learns to recover clean task predictions from x_t. Specifically, the TCTIA module integrates the noisy latent representation with shared image features and initial task predictions. Timestep information is incorporated into the attention process to adapt feature interaction at different noise levels. The resulting task-specific features are subsequently refined by denoising decoders equipped with time-conditioned residual convolutional blocks. Finally, the corresponding prediction heads generate the semantic segmentation, height estimation, and boundary detection outputs.

C. Label-Unified Encoding

The LuE module maps heterogeneous ground-truth labels from multiple tasks into a shared latent space, which provides a common target for the subsequent diffusion process.

Let the task labels associated with an input RS image be denoted as $\mathcal{Y} = \{y_k\}_{k=1}^K$. These labels differ in their value ranges, data types, and spatial structures. Directly combining them would therefore produce an inconsistent representation. To address this issue, task-specific encoders are first used to transform each label into a latent feature with the same spatial size and channel dimension. For categorical labels, such as semantic segmentation masks, the encoder is implemented using an MLP-based embedding layer. For dense spatial labels, such as height and boundary maps, a shallow convolutional encoder is used to aggregate local spatial information before projecting the label into the target embedding dimension. The task-specific encoding process is expressed as

$$z_k = E_k(y_k) \tag{4}$$

where $E_k(\cdot)$ denotes the encoder for the kth task, and $z_k \in \mathbb{R}^{H\times W\times C_{\text{embed}}}$ is the resulting task representation.

After obtaining the task-specific embeddings $\{z_k\}_{k=1}^K$, they are concatenated and fused into a clean shared task-label latent

$$x_0 = E_{\text{fuse}}(\text{Cat}(z_1, \ldots, z_K)) \tag{5}$$

where $E_{\text{fuse}}(\cdot)$ denotes the label fusion layer and x_0 represents the clean latent of all task labels. The task-specific encoders preserve the individual characteristics of different labels, while the fusion layer aligns them in a common latent space. The forward diffusion process is applied only to x_0. During training, a diffusion time t is randomly sampled, and Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is added according to the adopted variance schedule

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon \tag{6}$$

where x_t is the noisy representation at timestep t, β_t is the predefined variance, $\alpha_t = 1-\beta_t$, and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ denotes the cumulative product of α_t. It should be noted that the shared image features F_{shared} and the initial multitask predictions P_{init} are not involved in the forward noising process. They are used as image-derived conditions in the reverse process.

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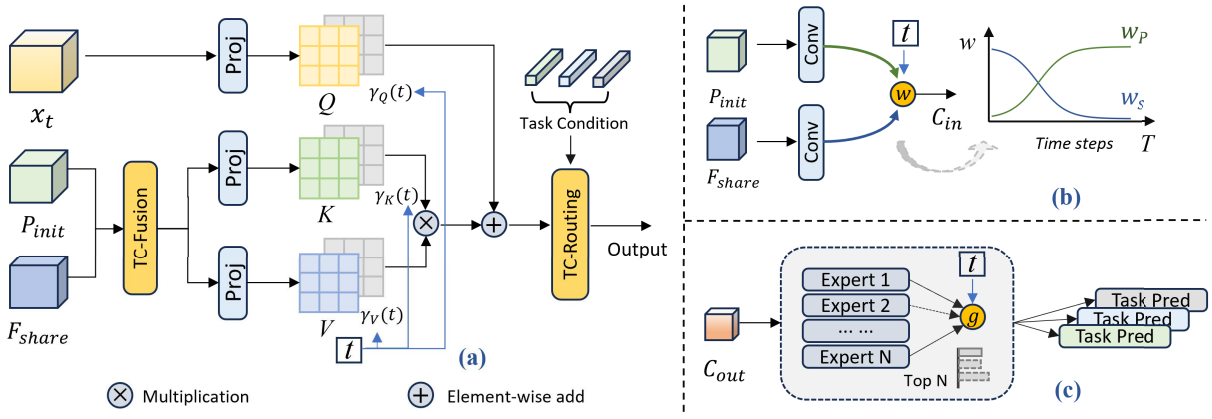


Fig. 3. (a) Architecture of TCTIA module. The shared image features F_{shared} and initial task predictions P_{init} are used as conditions to interact with the noisy latent representation x_t. (b) TC-Fusion: Time-conditioned fusion strategy to adaptively combine shared features and initial predictions based on the current diffusion timestep t. (c) TC-Routing: Time-conditioned routing layer to adaptively route the shared representation to different task heads.

This separation keeps the forward diffusion formulation well-defined while allowing the denoising decoder to exploit both image features and initial task predictions.

Unlike methods that construct an independent diffusion process for each task, the proposed module represents all task labels using a single shared latent. This design provides a unified diffusion target while retaining task-specific label characteristics through the individual encoders. It also enables the subsequent decoder to model interactions among semantic segmentation, height estimation, and boundary detection within the same denoising process.

D. Task-Interactive Decoding

The task-interactive decoding module is designed to reverse the diffusion process by iteratively denoising the noisy latent feature map x_t. As shown in Fig. 3, it consists of two main components: a TCTIA module and task-specific denoising decoders. The TCTIA module enhances the interactions between task-specific features and the noisy latent representation, while the denoising decoders progressively refine the denoised representations to produce final task outputs.

a) Time-conditioned task-interactive attention: To effectively aggregate features of different stages, both shared features F_{shared} and initial prediction P_{init} are used. Unlike current methods that simply concatenate these features, a dynamic fusion strategy is proposed to adaptively combine them based on the current diffusion timestep t. Specifically, the signal-to-noise ratio (SNR) at timestep t is transformed into its logarithmic form to guide the fusion process. The logSNR is defined as

$$\text{logSNR}(t) = \log\left(\frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}\right). \tag{7}$$

Then the weighting coefficients $w_S(t)$ and $w_P(t)$ are computed as

$$w_S(t) = \sigma\left(\text{logSNR}(t)\right) \tag{8}$$

$$w_P(t) = 1 - w_S(t) \tag{9}$$

where $\sigma(\cdot)$ is the sigmoid function to normalize the logSNR to $[0, 1]$. As shown in Fig. 3(b), weighting coefficients $w_S(t)$ and $w_P(t)$ vary with the timestep t, which allows the model to adaptively adjust the contributions of shared features and initial predictions during different denoising stages. The fused composite representation C_{in} is computed as

$$C_{\text{in}} = w_S(t) \cdot F_{\text{shared}} + w_P(t) \cdot P_{\text{init}}. \tag{10}$$

The motivation behind this design is that, at the early stages of the diffusion process (i.e., large t), the model relies more on the initial predictions to guide denoising, while at later stages (i.e., small t), it focuses more on high-level representations (shared features) to refine details. With the logSNR guiding the fusion, the model can dynamically adjust the contributions of different features according to the denoising stage.

As illustrated in Fig. 3, the TCTIA module enhances the interaction between the task-specific features and the noisy latent representation by performing attention, while incorporating timestep information into the attention process. Specifically, the composite representation C_{in} is used as both the key and value, while the noisy latent feature map x_t acts as the query input. Before the attention computation, a learnable embedding of the current timestep t is added to the query, key, and value features to guide the attention mechanism according to the denoising step. Then, all feature maps are projected into a common latent space by 1×1 convolutional layers, which are written as

$$Q = W_Q(x_t) + \gamma_Q(t) \tag{11}$$

$$K = W_K(C_{\text{in}}) + \gamma_K(t) \tag{12}$$

$$V = W_V(C_{\text{in}}) + \gamma_V(t) \tag{13}$$

where $\gamma(t)$ represents the embedded representation of timestep t. Then, the output is computed via an attention mechanism

$$C_{out} = Q + \text{Attention}(Q, K, V) \tag{14}$$

where Attention$(\cdot)$ denotes the standard scaled dot-product attention operation. The residual connection is employed to facilitate gradient flow and stabilize training.

The generated feature C_{out} is then used for task-specific interpretation. To efficiently aggregate task-specific features, a time-conditioned routing layer is introduced to adaptively route the shared representation to different task heads. This allows the model to leverage a shared set of expert networks while enabling task-specific feature extraction. To be specific,

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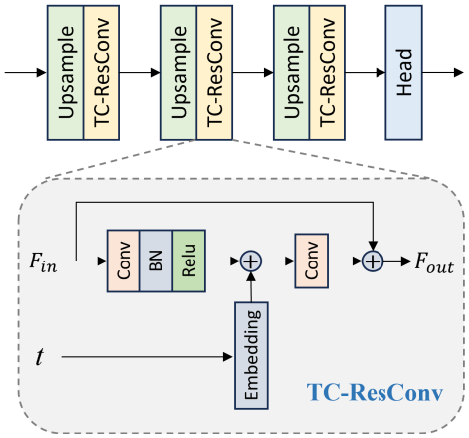


Fig. 4. Architecture of the TC-ResConv block.

let $\{G_i\}_{i=1}^{N}$ be a bank of lightweight experts on top of C_{out}. For task k at timestep t, a small router takes the feature C_{out} concatenated with the task embedding c_k and the timestep embedding $\gamma(t)$, and the task-specific feature is produced as a weighted combination of the experts

$$F^k(t) = \sum_{i=1}^{N} \tilde{g}_i^k(t)\, G_i\left(C_{out}, c_k, \gamma(t)\right) \tag{15}$$

where $\tilde{g}_i^k(t)$ is the normalized weight for expert G_i given by the router. This design enables task- and time-conditioned selection while sharing computation across tasks. The process is illustrated in Fig. 3(c). Then, the task-specific features $\{F^k(t)\}_{k=1}^{K}$ are fed into their respective denoising decoders for further refinement.

b) Denoising decoder: To further restore the clean task-specific outputs from noisy latent features, the denoising decoder is applied. Specifically, each decoder receives the output from the TCTIA module, and refines it progressively through several hierarchical decoding stages.

As shown in Fig. 4, the denoising decoder is composed of several time-conditioned residual convolution (TC-ResConv) blocks, upsampling operations, and prediction heads. The TC-ResConv block is designed to adaptively modulate feature transformation based on the current diffusion timestep t, and the upsampling operation is used to increase the spatial resolution of the feature maps. Finally, the task-specific prediction head is applied to generate the final output for each task.

Different from traditional residual convolutional blocks, the TC-ResConv block incorporates timestep information to guide the denoising process. Let $\gamma(t)$ denote a timestep-dependent modulation vector derived from the embedded timestep t through a learnable MLP. The output of the block is defined as

$$F_{\text{out}}(t) = F_{\text{in}} + \text{Conv}\left(\text{ConvBNReLu}\left(F_{\text{in}}\right) + \gamma(t)\right) \tag{16}$$

where ConvBNReLu$(\cdot)$ denotes a 3×3 convolution followed by normalization and ReLU activation. By conditioning the convolutional response on the diffusion timestep, TC-ResConv enables the decoder to adjust its transformation dynamically at each denoising stage. The upsampling operation is performed using bilinear interpolation, which is intended to enhance the spatial resolution of the feature maps. The upsampled feature map is then processed by the TC-ResConv block to produce a refined map. Finally, after several TC-ResConv blocks, the final feature map is obtained and fed into a task-specific prediction head to generate the final results.

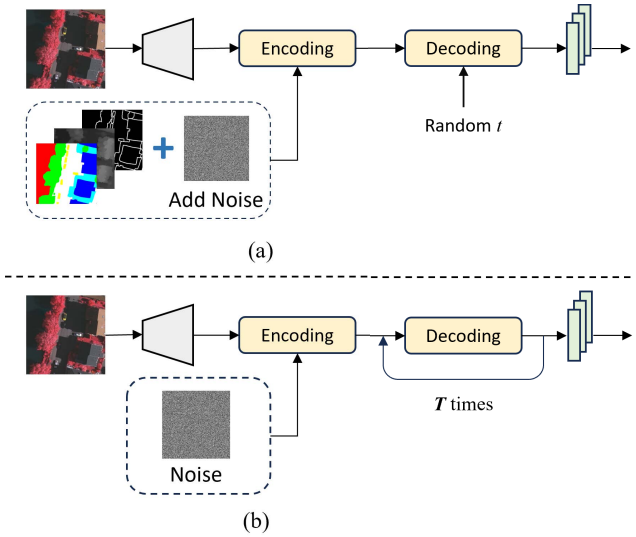


Fig. 5. Training and inference process of the proposed DiffRSMT. (a) Training phase. (b) Inference phase.

E. Model Optimization

The training and inference procedures of DiffRSMT are illustrated in Fig. 5. During training, the ground-truth task labels are encoded into the clean shared task-label latent x_0. The forward diffusion process is defined over T noise levels. Specifically, we adopt a linear variance schedule, in which the variance β_t increases from $\beta_{\min}$ to $\beta_{\max}$ over the diffusion horizon

$$\beta_t = \beta_{\min} + \frac{t-1}{T-1}\left(\beta_{\max} - \beta_{\min}\right), \qquad t = 1, \ldots, T. \tag{17}$$

At each training iteration, a timestep t is randomly sampled from $\{1, \ldots, T\}$, and the corresponding noisy latent x_t is generated from x_0 according to the forward diffusion process. At the final training timestep, the cumulative signal coefficient is $\bar{\alpha}_T = \prod_{t=1}^{T}(1-\beta_t)$. Under the adopted schedule, $\bar{\alpha}_T$ becomes sufficiently small, such that the contribution of the clean latent is largely suppressed and the Gaussian noise component dominates x_T. This makes the highest-noise training latent approximately consistent with the Gaussian initialization used during inference.

During inference, the reverse process is initialized with Gaussian noise and conditioned on the shared image features and initial task predictions extracted from the input image. Instead of traversing all T noise levels, we employ denoising diffusion implicit models (DDIMs) [35] sampling to skip intermediate states and perform the reverse process with a small number of denoising steps. At each sampling timestep t, the task-interactive decoder and prediction heads estimate the clean task outputs $\{\hat{y}_{\text{seg}}^t, \hat{y}_{\text{hei}}^t, \hat{y}_{\text{bou}}^t\}$ from the current noisy latent x_t. Before re-encoding, the segmentation logits are converted into discrete class labels using argmax, the boundary logits are transformed into probability maps using the sigmoid function,

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and the height predictions retain their normalized continuous form. These processed task predictions are then fed into their corresponding label encoders in LuE and fused to obtain the clean latent estimate $\hat{x}_0^t$ at the current sampling timestep. Given the current noisy latent x_t and the estimated clean latent $\hat{x}_0^t$, the residual noise is computed as

$$\hat{\epsilon}_t = \frac{x_t - \sqrt{\bar{\alpha}_t}\,\hat{x}_0^t}{\sqrt{1-\bar{\alpha}_t}} \tag{18}$$

where $\bar{\alpha}_t$ denotes the cumulative signal coefficient corresponding to timestep t. For the next sampling timestep $t' < t$, the next latent state is updated as

$$x_{t'} = \sqrt{\bar{\alpha}_{t'}}\,\hat{x}_0^t + \sqrt{1-\bar{\alpha}_{t'}}\,\hat{\epsilon}_t. \tag{19}$$

The updated latent $x_{t'}$ is then used as the input to the next denoising step, with no additional random noise introduced during the update. This procedure is repeated until the final task predictions are obtained.

Instead of a noise prediction objective, DiffRSMT adopts an output prediction objective, which is also commonly used in diffusion-based image prediction tasks. Previous diffusion-based methods show that task-specific discriminative losses are more suitable for image prediction than the standard L_2 objective used for noise or latent reconstruction [36]. Following this setting and the recent diffusion-based MTL framework [33], the reverse process of DiffRSMT is supervised through the final task predictions. This supervision directly matches the output form and optimization target of each task. Cross-entropy loss is used for semantic segmentation, mean squared error (mse) loss is used for height estimation, and binary cross-entropy loss is used for boundary detection. These losses match the output form and optimization objective of each task. The total loss is defined as a weighted sum of the three task losses

$$\begin{aligned} \mathcal{L}_{\text{total}} &= \lambda_{\text{seg}}\mathcal{L}_{\text{seg}} + \lambda_{\text{hei}}\mathcal{L}_{\text{hei}} + \lambda_{\text{bou}}\mathcal{L}_{\text{bou}} \\ \text{s.t.} \quad & \lambda_{\text{seg}} + \lambda_{\text{hei}} + \lambda_{\text{bou}} = 1 \end{aligned} \tag{20}$$

where $\mathcal{L}_{\text{seg}}$, $\mathcal{L}_{\text{hei}}$, and $\mathcal{L}_{\text{bou}}$ denote the losses for semantic segmentation, height estimation, and boundary detection, respectively. The coefficients λ_{seg}, λ_{hei}, and λ_{bou} balance the contributions of different tasks. Through these task-specific losses, the conditional denoising module and the prediction heads are optimized jointly in an end-to-end manner.

As a result, the proposed DiffRSMT is constructed as a unified framework for RS image multitask interpretation. The LuE module enables heterogeneous ground-truth labels to be embedded into a unified latent space. By leveraging a diffusion-based denoising process conditioned on shared image features and guided by a TCTIA mechanism, the model effectively captures both shared and task-interactive representations.

IV. Experimental Results

A. Datasets

To validate the effectiveness of the proposed DiffRSMT, extensive experiments are conducted on four publicly available RS image datasets. The details of these datasets are summarized as follows.

1) *Potsdam Dataset:* The dataset comprises 38 high-resolution RS images, each with a spatial size of 6000 × 6000 pixels. Each image contains four spectral bands, including red, green, blue, and near-infrared, together with the corresponding digital surface model (DSM). Semantic segmentation annotations are provided for six land-cover classes. In our experiments, images 4_10, 5_11, 2_11, 3_10, 6_11, and 7_12 are reserved for testing, while all remaining images are used for training. All images are uniformly cropped into nonoverlapping patches of 512 × 512 pixels.
2) *Vaihingen Dataset:* This dataset consists of 33 high-resolution RS images with different spatial sizes, with the largest image measuring approximately 2494 × 2064 pixels. Each image contains three spectral bands, including near-infrared, red, and green, together with the corresponding DSM. The dataset provides annotations for six semantic classes. In our experiments, images 5, 10, 15, 17, 21, 26, 30, 33, and 37 are reserved for testing, while all remaining images are used for training. All images are uniformly cropped into nonoverlapping patches of 512 × 512 pixels.
3) *GeoNRW Dataset:* The GeoNRW dataset covers the German state of North Rhine-Westphalia and includes RS images, DSM data, and segmentation maps annotated with ten land-cover classes. Each image has a spatial size of 512×512 pixels and contains four spectral bands: red, green, blue, and near-infrared. In our experiments, a total of 28 256 images are used for training, and 1940 images are used for testing. The dataset is publicly available as described in [47].
4) *QuebecTrees Dataset:* This dataset consists of high-resolution RS imagery collected over a temperate forest in Quebec, Canada [48]. Each image includes three spectral bands (red, green, and blue). A total of 23 000 individual tree crowns were manually segmented and labeled across 14 tree species classes. The imagery was cropped into patches of 512×512 pixels for training and evaluation. For this dataset, semantic segmentation and boundary detection tasks are considered, while height estimation is not applicable due to the absence of DSM data. In our experiments, 2626 images are used for training, and 420 images are used for testing.

For all datasets, boundary masks are generated from the semantic labels using a four-connected neighborhood comparison. A pixel is marked as a boundary if at least one adjacent pixel has a different class label, resulting in a two-pixel-wide binary boundary map without additional postprocessing. Pixels assigned an ignore label are excluded from boundary-mask generation.

B. Evaluation Metrics

To evaluate the performance of the proposed DiffRSMT on different tasks, we employ several standard metrics commonly used in RS image interpretation. The evaluation metrics are summarized as follows.

For semantic segmentation, the $F1$, intersection over union (IoU), and overall accuracy (OA) are adopted. These metrics

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are defined as follows:

$$F1 = \frac{2TP}{2TP + FP + FN} \tag{21}$$

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{22}$$

$$\text{OA} = \frac{TP + TN}{TP + TN + FP + FN} \tag{23}$$

where TP, TN, FP, and FN represent the true-positive, true-negative, false-positive, and false-negative pixel counts, respectively. For multiclass semantic segmentation, $F1$ and IoU are computed separately for each class and then macroaveraged over all evaluated classes. Pixels with ignored or unlabeled annotations are excluded from the confusion matrix and do not contribute to the calculation of $F1$, IoU, or OA.

For height estimation, the ground-truth height values are normalized to $[0, 1]$ before training and evaluation. The mse and mean absolute error (MAE) are calculated in the normalized height space as follows:

$$\text{mse} = \frac{1}{N}\sum_{i=1}^{N}(p_i - \hat{p}_i)^2 \tag{24}$$

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|p_i - \hat{p}_i| \tag{25}$$

where N is the number of pixels, p_i is the ground-truth height value, and $\hat{p}_i$ is the predicted height value for pixel i. For clearer presentation, the reported MAE and mse values are multiplied by 10^3 and are therefore shown with the symbol ‰ in the tables.

For boundary detection, the odsF [49] is used to evaluate the model's performance, which is a boundary detection metric that combines precision and recall to provide a single score for boundary detection performance assessment.

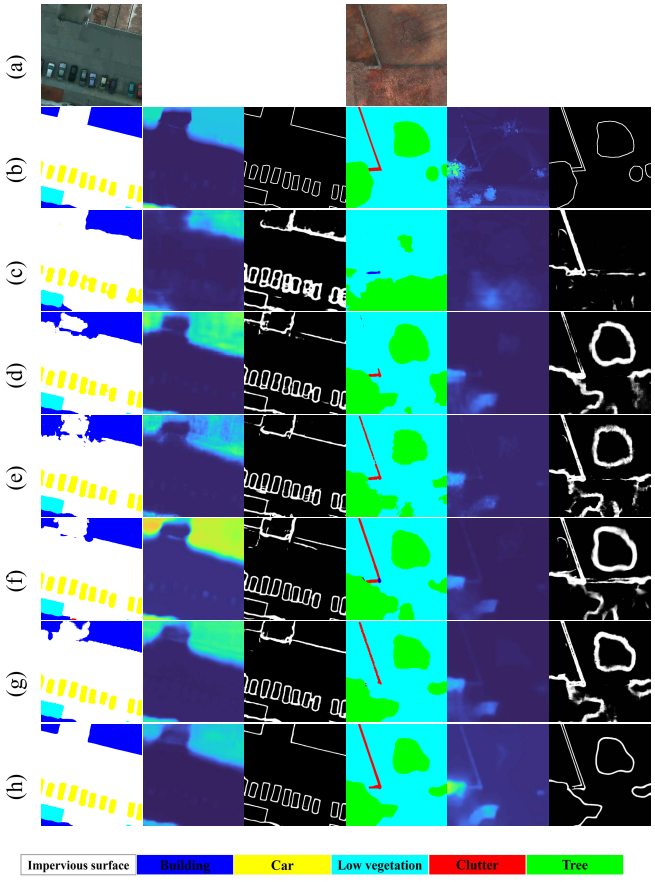


Fig. 6. MTL results over the Potsdam dataset. Each column corresponds to a specific task: semantic segmentation, height estimation, and boundary detection. Each row shows the results from a different MTL method. (a) Input image, (b) ground truth, (c) MTP, (d) BAMTL, (e) RSCoTr, (f) MCTNet, (g) MTMamba, and (h) proposed DiffRSMT.

C. Implementation Details

The proposed DiffRSMT is implemented using PyTorch and trained on a single NVIDIA RTX 4090D GPU. All input patches have a spatial size of 512×512 pixels, and the input image values are normalized to the range of $[0, 1]$. Standard data augmentations, including random flipping and cropping, are applied during training. ConvNeXt-Tiny pretrained on ImageNet is used as the backbone. The model is optimized using AdamW with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-5}. The batch size is set to 8, and the model is trained for 50 epochs. For the forward diffusion process, the diffusion horizon is set to $T = 1000$, with a linear variance schedule ranging from $\beta_{\text{min}} = 1 \times 10^{-4}$ to $\beta_{\text{max}} = 2 \times 10^{-2}$. During inference, three DDIM sampling steps are used to perform the reverse denoising process. The embedding dimension of the LuE module is set to 64, and the task-interactive attention mechanism uses a feature dimension of 128. The TC-ResConv blocks in the denoising decoder use a 3×3 kernel with a feature dimension of 128. Bilinear interpolation with a scale factor of 2 is used for upsampling.

D. Analysis of Results

1) Comparison Methods: The proposed DiffRSMT is compared with several state-of-the-art MTL methods, including MTFCN [43], BAMTL [44], RSCoTr [12], MCTNet [45], MTP [18], MTMamba [46], and TaskDiffusion [33]. MTFCN is a CNN-based MTL model that employs a fully convolutional network architecture, which is initially designed for tree species classification. BAMTL is a boundary-aware MTL model that incorporates boundary information into the feature extraction process. RSCoTr is a co-training Transformer-based MTL model. MCTNet is a multitask CNN and Transformer-integrated network. MTP is a multitask pretraining model that utilizes a pretrained foundation model. MTMamba is a multitask model that combines Mamba and CNN structures. TaskDiffusion is a diffusion-based MTL model that employs a conditional diffusion decoder to model interactions among multiple dense prediction tasks. Apart from the above MTL methods, several state-of-the-art single- and dual-task methods are also used for comparison, including MSGCNet [37], UNetFormer [38], RS3Mamba [39], HeightFormer [40], SCENet [42], and FAUNet [41]. MSGCNet, UNetFormer, and RS3Mamba are state-of-the-art semantic segmentation models. HeightFormer is a height estimation model, FAUNet is a boundary detection model, and SCENet is a dual-task model that jointly performs semantic segmentation and height estimation.

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TABLE I

QUANTITATIVE EVALUATION OF DIFFERENT MTL METHODS OVER THE POTSDAM DATASET. THE VALUES ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER TEN RUNS

Task	Method	Semantic Segmentation			Height Estimation		Boundary Detection
		F1(%)↑	mIoU(%)↑	OA(%)↑	MSE(‰)↓	MAE(‰)↓	odsF(%)↑
Segmentation	MSGCNet [37]	84.95 ± 0.42	74.91 ± 0.68	88.16 ± 0.24	-	-	-
	UNetFormer [38]	84.85 ± 0.38	74.76 ± 0.72	88.13 ± 0.21	-	-	-
	RS3Mamba [39]	85.18 ± 0.31	75.22 ± 0.56	88.34 ± 0.18	-	-	-
Height	HeightFormer [40]	-	-	-	6.95 ± 0.22	**42.43** ± 0.31	-
Boundary	FAUNet [41]	-	-	-	-	-	43.21 ± 0.79
Dual-task	SCENet [42]	84.87 ± 0.49	74.84 ± 0.83	88.29 ± 0.27	7.12 ± 0.15	43.85 ± 0.24	-
Multi-task	MTP [18]	73.56 ± 0.74	59.35 ± 1.38	77.72 ± 0.82	14.28 ± 0.31	68.29 ± 0.34	28.86 ± 0.76
	MTFCN [43]	75.17 ± 0.68	60.14 ± 1.21	78.65 ± 0.71	13.32 ± 0.27	54.24 ± 0.30	31.23 ± 0.69
	BAMTL [44]	83.99 ± 0.46	73.53 ± 0.79	87.25 ± 0.33	8.03 ± 0.18	48.31 ± 0.25	45.41 ± 0.55
	RSCoTr [12]	83.68 ± 0.63	72.75 ± 1.12	86.63 ± 0.58	15.25 ± 0.29	71.14 ± 0.33	44.98 ± 0.64
	MCTNet [45]	84.68 ± 0.39	74.50 ± 0.65	87.48 ± 0.29	7.14 ± 0.14	44.53 ± 0.22	46.52 ± 0.47
	MTMamba [46]	84.33 ± 1.16	74.22 ± 1.10	88.07 ± 0.98	7.28 ± 0.66	45.14 ± 0.64	47.54 ± 0.73
	TaskDiffusion [33]	84.75 ± 0.57	74.51 ± 0.91	87.45 ± 0.46	7.31 ± 0.19	44.55 ± 0.27	46.42 ± 0.51
	DiffRSMT (Ours)	**85.58** ± 0.28	**75.83** ± 0.44	**88.91** ± 0.16	**6.74** ± 0.10	43.33 ± 0.18	**48.12** ± 0.36

TABLE II

QUANTITATIVE EVALUATION OF DIFFERENT MTL METHODS OVER THE VAIHINGEN DATASET. THE VALUES ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER TEN RUNS

Task	Method	Semantic Segmentation			Height Estimation		Boundary Detection
		F1(%)↑	mIoU(%)↑	OA(%)↑	MSE(‰)↓	MAE(‰)↓	odsF(%)↑
Segmentation	MSGCNet [37]	84.87 ± 0.52	74.22 ± 0.94	88.36 ± 0.31	-	-	-
	UNetFormer [38]	87.06 ± 0.41	77.37 ± 0.71	88.46 ± 0.24	-	-	-
	RS3Mamba [39]	87.86 ± 0.35	78.46 ± 0.58	88.74 ± 0.28	-	-	-
Height	HeightFormer [40]	-	-	-	26.94 ± 0.21	137.09 ± 0.38	-
Boundary	FAUNet [41]	-	-	-	-	-	58.73 ± 0.65
Dual-task	SCENet [42]	84.59 ± 0.63	73.86 ± 1.08	88.18 ± 0.47	29.08 ± 0.32	144.40 ± 0.45	-
Multi-task	MTP [18]	76.82 ± 0.82	63.82 ± 1.55	84.66 ± 1.18	34.71 ± 0.51	149.90 ± 0.56	55.95 ± 0.87
	MTFCN [43]	77.34 ± 0.76	65.24 ± 1.42	85.45 ± 1.01	33.12 ± 0.47	147.23 ± 0.52	56.28 ± 0.81
	BAMTL [44]	86.92 ± 0.49	77.15 ± 0.83	88.34 ± 0.36	27.59 ± 0.28	136.54 ± 0.41	68.32 ± 0.56
	RSCoTr [12]	85.08 ± 0.68	75.86 ± 1.19	86.09 ± 0.79	28.05 ± 0.34	148.19 ± 0.50	53.78 ± 0.74
	MCTNet [45]	87.25 ± 0.43	77.63 ± 0.69	88.73 ± 0.27	27.48 ± 0.25	132.53 ± 0.36	68.41 ± 0.44
	MTMamba [46]	86.68 ± 1.17	76.82 ± 1.06	88.58 ± 1.02	36.22 ± 0.92	156.30 ± 0.64	67.09 ± 0.71
	TaskDiffusion [33]	87.18 ± 0.57	76.93 ± 0.96	88.31 ± 0.42	36.74 ± 0.39	161.28 ± 0.54	67.19 ± 0.61
	DiffRSMT (Ours)	**88.14** ± 0.29	**78.81** ± 0.44	**88.97** ± 0.22	**26.85** ± 0.21	**132.12** ± 0.19	**68.74** ± 0.25

For comparison, all methods are evaluated using the same backbone, input modalities, dataset splits, data augmentation, and training settings. Specifically, the original backbone encoders are replaced by the same ConvNeXt-Tiny backbone, while the method-specific fusion, decoding, and prediction modules are all retained. This setting reduces the influence of differences in encoder capacity, allowing the comparison to focus more directly on the multitask modeling mechanisms. Therefore, the reported results should be interpreted as controlled comparisons under a unified backbone and training protocol. For Potsdam, Vaihingen, and GeoNRW, DSM data are used only as supervision for height estimation and are not provided as model inputs. Thus, all methods receive the same optical image inputs. For QuebecTrees, only semantic segmentation and boundary detection are evaluated, and no DSM data are involved.

2) Quantitative and Visual Results: Tables I–IV and Figs. 6–9 present the results of different MTL methods across four datasets: Potsdam, Vaihingen, GeoNRW, and QuebecTrees, covering three tasks: semantic segmentation, height estimation, and boundary detection. To account for variations caused by random initialization and training, all quantitative results are reported as the mean and standard deviation over ten independent runs.

Table I reports the results on the Potsdam dataset. DiffRSMT achieves the highest mean values for the three semantic segmentation metrics. Compared with the strongest single-task baseline, DiffRSMT improves mIoU and OA by 0.61 and 0.57 percentage points, respectively. For height estimation, DiffRSMT obtains the lowest mse, while HeightFormer achieves a slightly lower MAE. For boundary detection, DiffRSMT improves the odsF score by 0.58% over MTMamba. In addition, DiffRSMT generally presents lower or comparable standard deviations, indicating stable performance across repeated runs. As shown in Fig. 6, DiffRSMT produces more complete building and vehicle regions. Its

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TABLE III

QUANTITATIVE EVALUATION OF DIFFERENT MTL METHODS OVER THE GEONRW DATASET. THE VALUES ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER TEN RUNS

Task	Method	Semantic Segmentation			Height Estimation		Boundary Detection
		F1(%)↑	mIoU(%)↑	OA(%)↑	MSE(‰)↓	MAE(‰)↓	odsF(%)↑
Segmentation	MSGCNet [37]	75.12 ± 0.82	62.02 ± 1.03	81.05 ± 0.61	-	-	-
	UNetFormer [38]	74.60 ± 0.94	61.36 ± 1.22	80.29 ± 0.74	-	-	-
	RS3Mamba [39]	74.19 ± 0.88	60.83 ± 1.16	80.15 ± 0.68	-	-	-
Height	HeightFormer [40]	-	-	-	32.48 ± 0.46	141.85 ± 0.91	-
Boundary	FAUNet [41]	-	-	-	-	-	48.92 ± 0.76
Dual-task	SCENet [42]	73.50 ± 1.03	60.07 ± 1.34	80.33 ± 0.89	32.16 ± 0.72	142.52 ± 1.14	-
Multi-task	MTP [18]	66.30 ± 1.52	52.22 ± 1.83	74.67 ± 1.96	33.47 ± 1.86	146.06 ± 2.11	46.10 ± 1.15
	MTFCN [43]	68.51 ± 1.41	55.17 ± 1.71	75.86 ± 1.74	32.34 ± 1.62	145.01 ± 1.92	46.78 ± 1.08
	BAMTL [44]	73.28 ± 0.97	59.83 ± 1.25	79.23 ± 0.92	34.27 ± 0.84	145.27 ± 1.23	53.04 ± 0.71
	RSCoTr [12]	74.22 ± 1.18	60.14 ± 1.46	79.77 ± 1.21	35.71 ± 1.42	147.71 ± 1.81	52.98 ± 0.83
	MCTNet [45]	75.03 ± 0.79	61.95 ± 0.98	80.19 ± 0.66	33.84 ± 0.73	144.16 ± 1.07	55.14 ± 0.59
	MTMamba [46]	71.57 ± 1.24	57.89 ± 1.55	78.67 ± 1.38	37.93 ± 1.57	155.75 ± 2.04	54.94 ± 0.88
	TaskDiffusion [33]	71.17 ± 1.05	57.68 ± 1.11	78.45 ± 1.14	36.99 ± 1.65	151.70 ± 2.17	53.94 ± 0.78
	DiffRSMT (Ours)	**75.79** ± 0.55	**62.76** ± 0.72	**81.32** ± 0.41	**31.81** ± 0.39	**141.31** ± 0.68	**57.12** ± 0.47

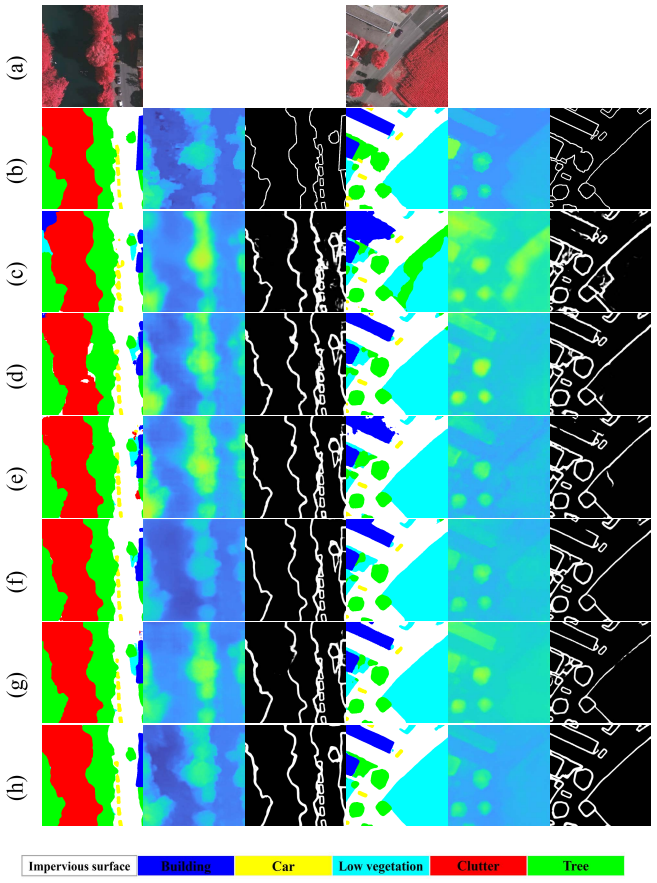


Fig. 7. MTL results over the Vaihingen dataset. Each column corresponds to a specific task: semantic segmentation, height estimation, and boundary detection. Each row shows the results from a different MTL method. (a) Input image, (b) ground truth, (c) MTP, (d) BAMTL, (e) RSCoTr, (f) MCTNet, (g) MTMamba, and (h) proposed DiffRSMT.

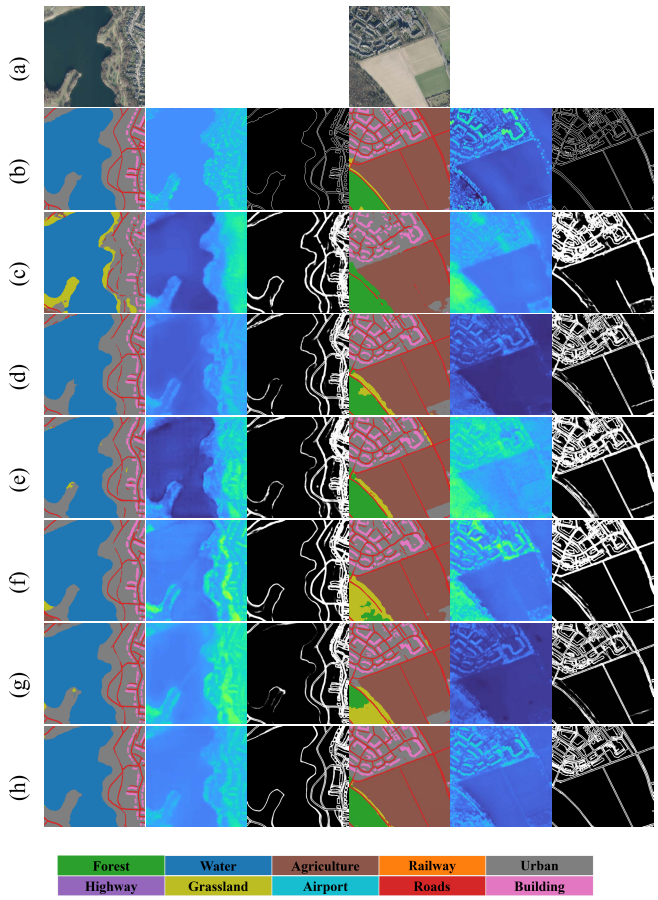


Fig. 8. MTL results over the GeoNRW dataset. Each column corresponds to a specific task: semantic segmentation, height estimation, and boundary detection. Each row shows the results from a different MTL method. (a) Input image, (b) ground truth, (c) MTP, (d) BAMTL, (e) RSCoTr, (f) MCTNet, (g) MTMamba, and (h) proposed DiffRSMT.

height predictions are spatially coherent, and the predicted boundaries are more closely aligned with the reference edges.

Table II presents the quantitative results of different MTL methods on the Vaihingen dataset. DiffRSMT achieves better performance across the three tasks. For semantic segmentation, it obtains the highest $F1$, mIoU, and OA scores. For height estimation, it yields the lowest mse and MAE, indicating more

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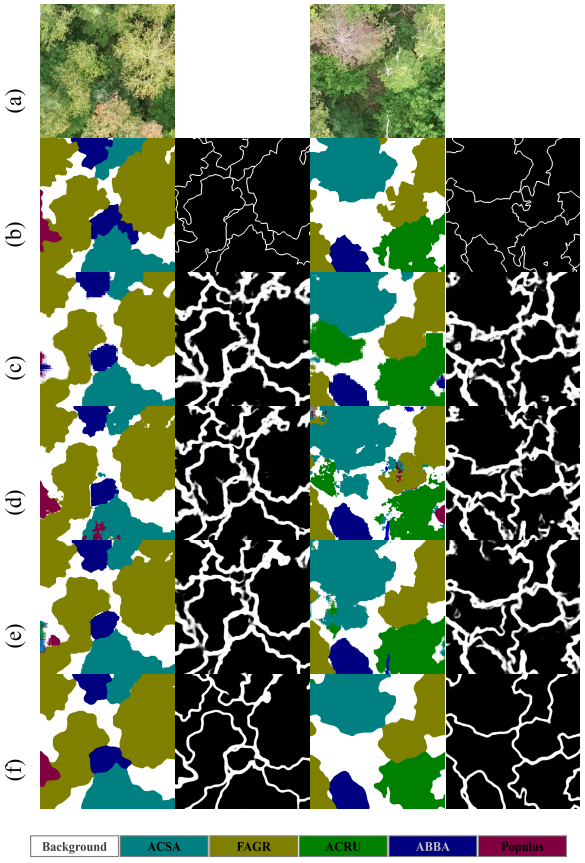


Fig. 9. MTL results over the QuebecTrees dataset. Each column corresponds to a specific task: semantic segmentation and boundary detection. Each row shows the results from a different MTL method. (a) Input image, (b) ground truth, (c) MTFCN, (d) MTP, (e) MTMamba, and (f) proposed DiffRSMT.

TABLE IV
QUANTITATIVE EVALUATION OF DIFFERENT MTL METHODS OVER THE QUEBECTREES DATASET. THE VALUES ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER TEN RUNS

Method	Segmentation			Boundary
	F1(%)↑	mIoU(%)↑	OA(%)↑	odsF(%)↑
SCENet [42]	71.25 ± 1.08	52.55 ± 1.31	73.79 ± 0.92	59.41 ± 0.83
MTP [18]	69.62 ± 1.62	50.13 ± 1.87	71.65 ± 1.71	52.15 ± 1.29
MTFCN [43]	72.08 ± 1.21	54.17 ± 1.46	74.31 ± 1.03	60.12 ± 0.71
BAMTL [44]	71.34 ± 1.34	53.04 ± 1.58	73.18 ± 1.19	54.46 ± 1.16
MCTNet [45]	70.32 ± 1.47	52.01 ± 1.72	73.50 ± 1.38	56.41 ± 0.98
MTMamba [46]	73.21 ± 0.86	55.18 ± 1.07	74.53 ± 0.74	58.73 ± 0.79
TaskDiffusion [33]	72.51 ± 0.96	54.83 ± 1.12	74.13 ± 0.61	56.93 ± 0.58
DiffRSMT (Ours)	**75.33** ± 0.61	**57.98** ± 0.68	**75.41** ± 0.43	**61.01** ± 0.52

accurate elevation prediction. It also achieves an odsF score of 68.74% for boundary detection, outperforming the comparison methods. Fig. 7 shows the corresponding visual results. DiffRSMT more clearly distinguishes trees from low vegetation and produces more continuous object boundaries, whereas several competing methods generate blurred or ambiguous predictions in these regions.

Table III shows the comparison results of different MTL methods on the GeoNRW dataset. DiffRSMT also achieves improved scores across all tasks, with an $F1$ of 75.79%, mIoU of 62.76%, and OA of 81.32% for semantic segmentation. For height estimation, DiffRSMT yields the lowest mse (31.81‰) and MAE (141.31‰), while also achieving the best boundary detection score (57.12%) and outperforming MCTNet by 1.98%. Fig. 8 presents the visual results of different MTL methods on the GeoNRW dataset. We can see that the proposed DiffRSMT outperforms other approaches in segmenting water bodies and roads, providing clearer boundaries and more reliable class predictions.

TABLE V
ABLATION STUDY OF THE LUE MODULE

Dataset	Ours (w/o LuE)	F1(%)↑	MSE(‰) ↓	odsF(%) ↑
Potsdam	+ Noise	81.82	10.32	40.25
	+ Conv	84.02	7.23	45.43
	+ LuE	**85.58**	**6.74**	**48.12**
Vaihingen	+ Noise	82.23	28.67	62.43
	+ Conv	85.87	27.73	66.62
	+ LuE	**88.14**	**26.85**	**68.74**
GeoNRW	+ Noise	70.16	33.45	52.88
	+ Conv	74.12	32.23	56.78
	+ LuE	**75.79**	**31.81**	**57.12**
QuebecTrees	+ Noise	71.68	-	57.31
	+ Conv	73.89	-	59.78
	+ LuE	**75.33**	**-**	**61.01**

As shown in Table IV, different MTL methods are evaluated on the QuebecTrees dataset for semantic segmentation and boundary detection. We can see that the proposed DiffRSMT outperforms all other methods across both tasks. Specifically, it achieves the highest $F1$-score (75.33%), mIoU (57.98%), and OA (75.41%) in semantic segmentation. Fig. 9 presents the visual comparison of different MTL methods on the QuebecTrees dataset. Owing to the complex characteristics of tree crowns, which often exhibit fine and irregular edge structures, accurate segmentation and precise boundary delineation remain highly challenging. Among all methods, the proposed DiffRSMT offers better performance in both segmentation and boundary detection.

Overall, the results show that DiffRSMT provides improved performance across multiple RS datasets. When compared with several MTL methods, DiffRSMT achieves a more balanced performance across semantic segmentation, height estimation, and boundary detection, while generally exhibiting lower variance across repeated runs. These results support the effectiveness of the proposed framework.

E. Ablation Study

To assess the effectiveness of different components in the proposed DiffRSMT, an ablation study on LuE, TCTIA, and TC-ResConv is conducted.

1) Label-Unified Encoding: Table V presents the ablation results for the LuE module. To evaluate its effectiveness, three variants are compared. In the "+ Noise" variant, the LuE module is removed, and the diffusion latent is initialized with Gaussian noise without any task-label embedding. In the "+ Conv" variant, the task labels are concatenated and projected into the latent space using convolutional layers with the same dimension as LuE, but without task-specific encoders

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TABLE VI
ABLATION STUDY OF THE TCTIA AND TC-RESCONV MODULES

Dataset	Method	F1(%)↑	MSE(‰) ↓	odsF(%) ↑
Potsdam	Baseline	82.37	8.13	45.72
	+ Only TCTIA	84.10	7.21	47.78
	+ Only TC-C	83.57	7.45	46.89
	+ TCTIA & TC-C	**85.58**	**6.74**	**48.12**
Vaihingen	Baseline	86.13	28.53	66.28
	+ Only TCTIA	87.64	27.05	67.94
	+ Only TC-C	87.02	27.91	68.05
	+ TCTIA & TC-C	**88.14**	**26.85**	**68.74**
GeoNRW	Baseline	73.17	33.71	55.01
	+ Only TCTIA	74.76	32.01	56.76
	+ Only TC-C	75.12	32.43	56.17
	+ TCTIA & TC-C	**75.79**	**31.81**	**57.12**
QuebecTrees	Baseline	73.23	-	59.07
	+ Only TCTIA	74.87	-	**61.35**
	+ Only TC-C	74.23	-	60.96
	+ TCTIA & TC-C	**75.33**	-	61.01

TABLE VII
ANALYSIS OF THE NONDIFFUSION COUNTERPART AND DIFFERENT NUMBERS OF DENOISING STEPS

Dataset	Metric	Non-Diff.	Step 1	Step 2	Step 3	Step 4	Step 5
Potsdam	F1(%)↑	82.07	84.20	85.02	85.58	**85.66**	85.51
	MSE(‰)↓	8.94	7.45	7.02	6.74	**6.68**	6.79
	odsF(%)↑	44.37	47.05	47.72	**48.12**	48.01	47.89
Vaihingen	F1(%)↑	85.13	87.10	87.72	88.14	**88.21**	88.05
	MSE(‰)↓	29.53	27.72	27.18	26.85	**26.77**	26.93
	odsF(%)↑	66.28	67.20	68.05	**68.74**	68.69	68.51
GeoNRW	F1(%)↑	72.88	74.32	75.02	**75.79**	75.74	75.61
	MSE(‰)↓	34.23	32.95	32.28	31.81	**31.70**	31.89
	odsF(%)↑	54.32	55.98	56.65	**57.12**	57.08	56.91
QuebecTrees	F1(%)↑	72.45	74.12	74.78	75.33	**75.40**	75.22
	odsF(%)↑	58.41	60.05	60.72	**61.01**	60.96	60.81

or the proposed label-unified fusion design. The "+ LuE" variant denotes the full model, in which each task label is first processed by its corresponding task-specific encoder and then fused into a shared latent representation. Across all datasets, incorporating LuE leads to better performance in all three tasks. For instance, on the Potsdam dataset, LuE improves the $F1$-score from 81.82% to 85.58% and the boundary detection score (odsF) from 40.25% to 48.12%. Similar improvements can be observed on the Vaihingen, GeoNRW, and QuebecTrees datasets. These results demonstrate the advantage of learning shared and unified representations that align heterogeneous task labels within a common embedding space.

2) TCTIA and TC-ResConv: Table VI presents the ablation results of the TCTIA module and the time-conditioned residual convolution (TC-ResConv) blocks. For compact presentation, TC-ResConv is abbreviated as TC-C in the table. Four configurations are evaluated: the baseline without either module, the model using only TCTIA, the model using only TC-ResConv, and the full model integrating both components. In the baseline, TCTIA is replaced with standard scaled dot-product cross-attention, where the noisy latent representation x_t serves as the query, and the concatenated shared image features F_{shared} and initial task predictions P_{init} serve as the key and value. Meanwhile, TC-ResConv is replaced with standard residual convolutional blocks that use the same feature dimension and convolution kernels, but without timestep-dependent modulation. For a fair comparison, all four configurations retain the same overall framework and training settings, so that the performance differences mainly reflect the effects of TCTIA and TC-ResConv.

From the table, we can find that both TCTIA and TC-ResConv contribute to improving the model's performance across all tasks. For example, on the Potsdam dataset, using TCTIA alone improves the $F1$-score to 84.10% and the boundary detection score to 47.78%. Using TC-ResConv alone achieves similar improvements. On the QuebecTrees dataset, the full configuration also achieves the highest $F1$-score of 75.33%, whereas the model using only TCTIA obtains a slightly higher odsF score. This result suggests a minor tradeoff between segmentation and boundary detection when TCTIA and TC-ResConv are jointly optimized on this dataset. Overall, the combined configuration provides more balanced improvements across datasets and tasks, supporting the complementary roles of the two components.

F. Analysis of Parameters

In this section, we investigate the influence of parameters on the model's performance, including the weighting coefficients for multitask losses and the number of denoising steps in the diffusion process. The analysis is presented as follows.

1) Analysis of the Weighting Coefficient: The weighting coefficients for MTL losses are crucial parameters for balancing the contributions of different tasks during training. To evaluate their influence, an experiment is conducted by varying the coefficients λ_{seg} and λ_{hei}, while keeping their sum equal to one. In this setup, λ_{seg} and λ_{hei} are varied explicitly, while λ_{bou} is derived as $1 - \lambda_{\text{seg}} - \lambda_{\text{hei}}$. A comprehensive index Δm [50], which was proposed for evaluating MTL performance, is used to assess the overall performance of our proposed method. As shown in Fig. 10, the x- and y-axes represent the values of λ_{bou} and λ_{seg}, respectively, and the heatmap values indicate the performance of the model with different weight combinations. We can observe that the model's performance varies significantly with different combinations of coefficients. For instance, when λ_{seg} and λ_{bou} are set to 0.05 (thus λ_{hei} is set to 0.9), the model provides poor performance as the tasks are unbalanced. In contrast, when the coefficients are set in a balanced manner (near 0.35 in the figure), the model achieves the best performance. Therefore, for the Potsdam, Vaihingen, and GeoNRW datasets, we set the three weighting coefficients to 0.33, 0.33, and 0.34, respectively.

For the QuebecTrees dataset, the model is designed to focus more on semantic segmentation and boundary detection tasks. As shown in Fig. 10(d), the best performance is achieved when λ_{seg} and λ_{bou} are both set to 0.5, when the model focuses equally on semantic segmentation and boundary detection tasks.

2) Analysis of Denoising Steps: Table VII compares the nondiffusion counterpart with DiffRSMT using one to five denoising steps. The nondiffusion model shares the same backbone, task-interactive module, decoder, and prediction heads, but directly predicts outputs without forward noising

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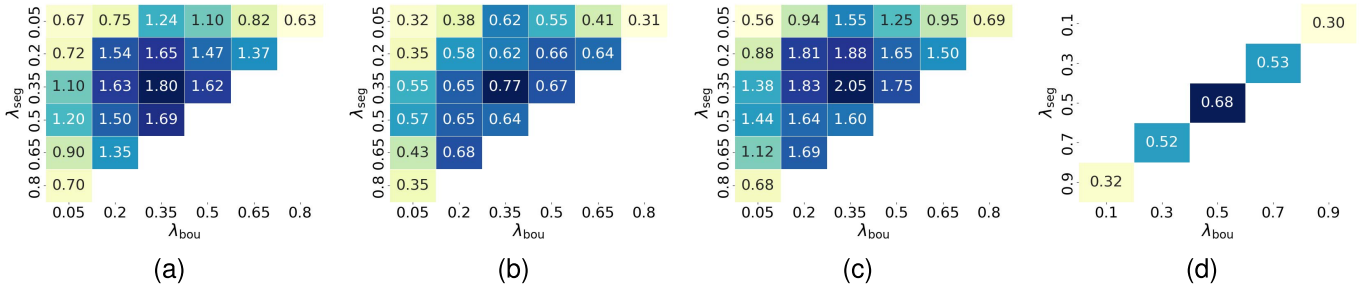


Fig. 10. Ablation study of the weighting coefficients of DiffRSMT over four datasets. Each subfigure shows the heatmap of Δm values for different combinations of λ_{seg}, λ_{hei}, and λ_{bou}.(a) Potsdam dataset. (b) Vaihingen dataset. (c) GeoNRW dataset. (d) QuebecTrees dataset.

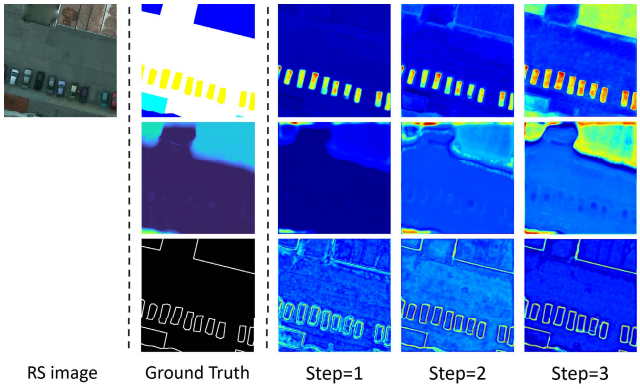


Fig. 11. Visualization of feature maps corresponding to different denoising steps.

TABLE VIII
COMPUTATIONAL COST (FLOPS), MODEL SIZE (MB), MEMORY USAGE (MB), AND INFERENCE SPEED (MS/IMG) OF DIFFERENT MTL APPROACHES

Method	FLOPs (G)	Model Size (MB)	Memory Usage (MB)	Infer. Time (ms/img)
BAMTL	**188.7**	60.58	**653.4**	**26.17**
MTP	239.0	95.38	816.3	27.64
RSCoTr	221.3	78.23	785.6	28.93
MCTNet	214.7	75.89	775.6	29.45
MTMamba	201.1	71.50	758.6	41.45
DiffRSMT(Ours)	225.1	**52.93**	764.3	31.25

or iterative denoising. The table indicates that introducing a diffusion model provides better performance when compared with the nondiffusion model across all four datasets. For example, on Potsdam, one denoising step raises the $F1$-score from 82.07% to 84.20% and the odsF score from 44.37% to 47.05%. Performance generally increases as the number of denoising steps increases from one to three. Steps 3 and 4 achieve the best results on different metrics, indicating that additional denoising can provide limited further refinement. However, increasing the number of steps to five leads to a slight decrease across the evaluated datasets. Therefore, to balance the overall performance and computational cost, the number of denoising steps is set to 3 in our experiments. This is reasonable because the diffusion process is operated in a compact label-unified latent space rather than directly in the high-dimensional image space. In addition, the denoising process is strongly conditioned on the shared image features and initial task predictions, so each step mainly refines a constrained task representation instead of reconstructing the outputs from unconstrained noise.

To further illustrate the denoising process, feature maps at different denoising steps are shown in Fig. 11. These feature maps are extracted from the last convolutional layer of the decoder. As the denoising step increases, the feature maps become progressively clearer and more structured, indicating that the iterative refinement process effectively enhances the model's ability to capture fine-grained details and improve overall prediction quality. Moreover, the visual results demonstrate that the model gradually learns to emphasize relevant features while suppressing noise across multiple denoising steps, leading to improved performance for all tasks.

G. Computational Cost and Memory Usage

Table VIII presents the computational cost, model size, memory usage, and inference speed of different MTL methods. The memory usage is measured as the peak GPU memory allocated during inference in the PyTorch environment, with a batch size of 1 and an input size of 512×512 for all methods. From the table, it can be seen that the proposed DiffRSMT does not achieve the lowest FLOPs, memory usage, or inference time, mainly because several denoising steps are required to progressively refine the predictions. The main efficiency advantage of DiffRSMT lies in its compact model size. It requires only 52.93 MB, which is smaller than those of the compared methods. For example, BAMTL achieves the lowest FLOPs (188.7G) and memory usage (653.4 MB), but has a larger model size of 60.58 MB. In terms of inference speed, DiffRSMT requires 31.25 ms/img, which remains comparable to the other methods. Overall, these results indicate that DiffRSMT provides a reasonable tradeoff between predictive performance and computational efficiency.

V. CONCLUSION

In this article, we proposed DiffRSMT, a diffusion-driven MTL framework for joint RS image segmentation, height estimation, and boundary detection. DiffRSMT unifies heterogeneous task labels in a shared latent space and models task interactions through time-conditioned attention and convolution during denoising. Experiments on four public RS datasets

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demonstrate that DiffRSMT provides a better overall performance than existing MTL methods. However, the iterative denoising process introduces additional computational cost and inference time compared with direct prediction models. In future work, we will investigate more efficient diffusion strategies to reduce this overhead and improve practical deployment.

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